AUTONOMOUS CYBER DEFENDER

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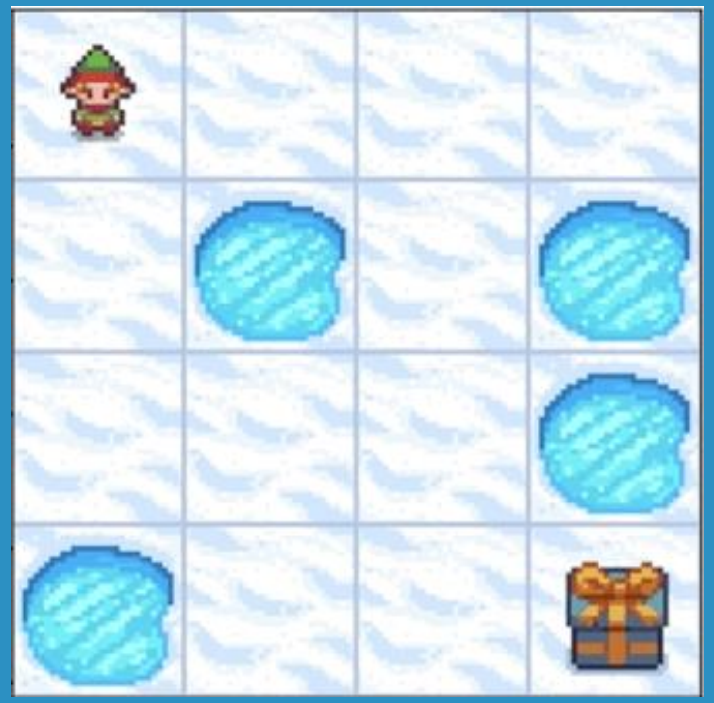
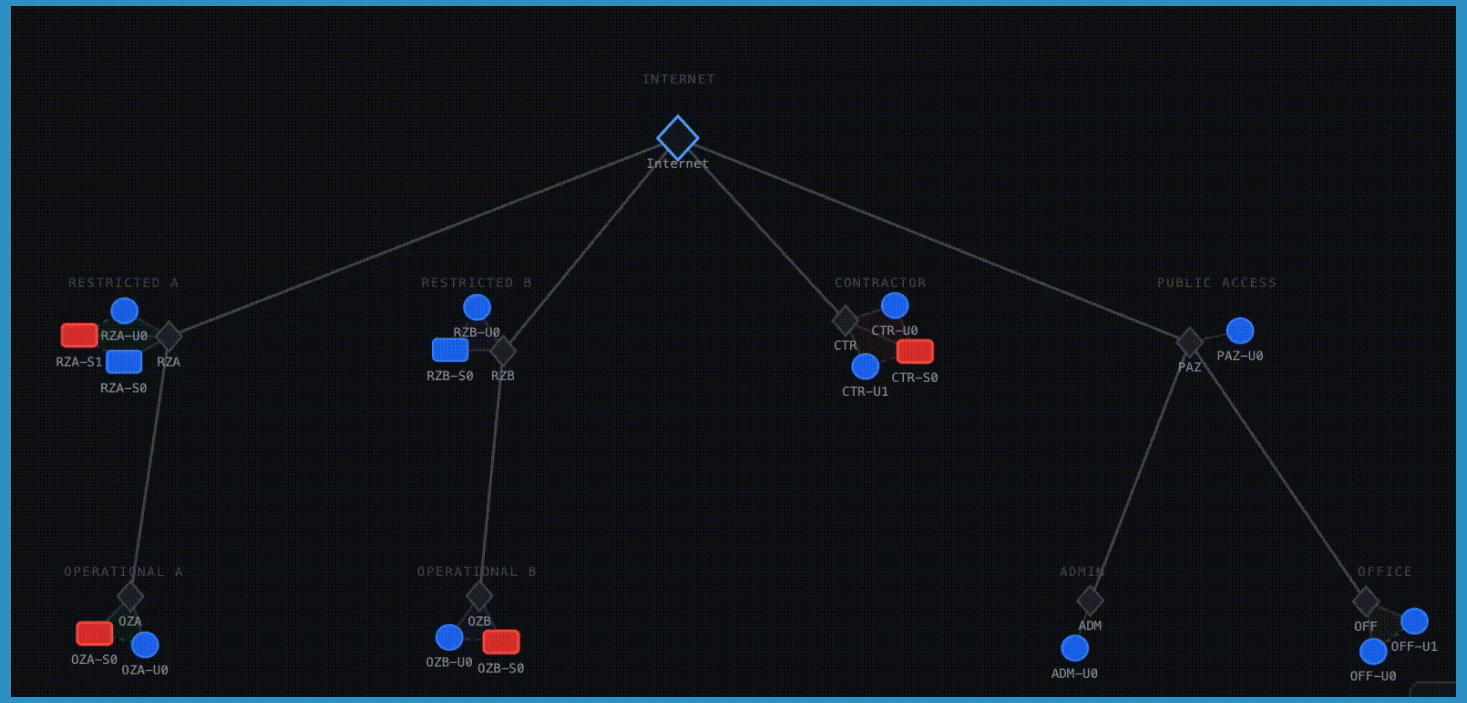
Simulation to Reality

Christmas

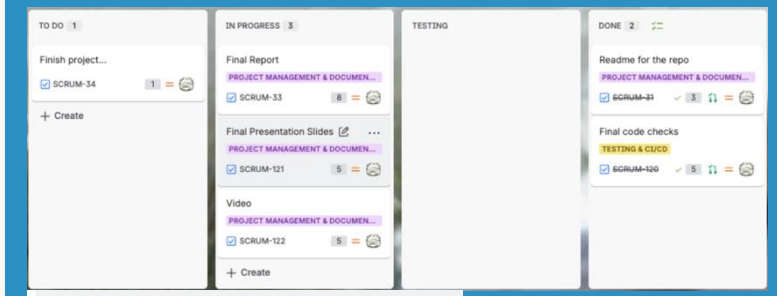


Summer



PROCESS

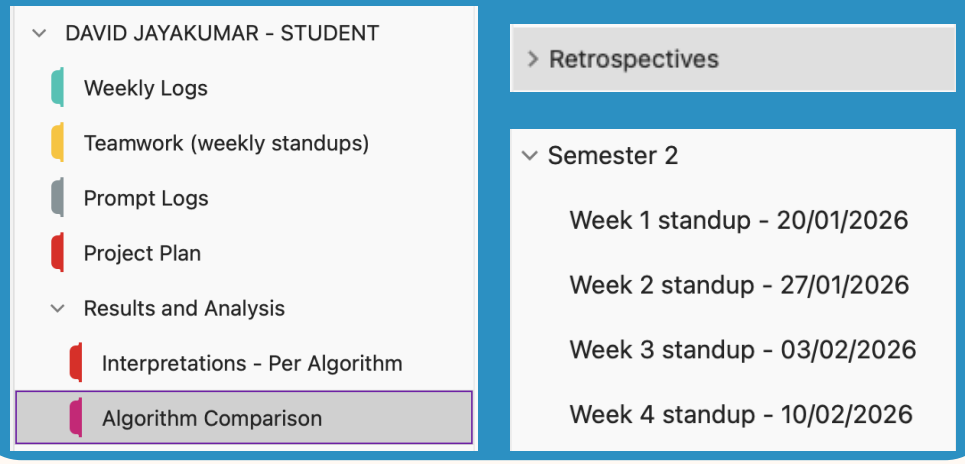
JIRA



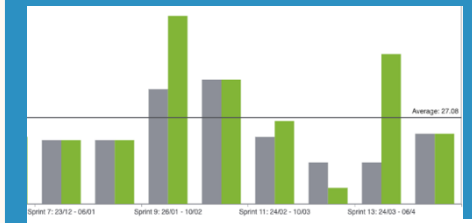
Deliverables:

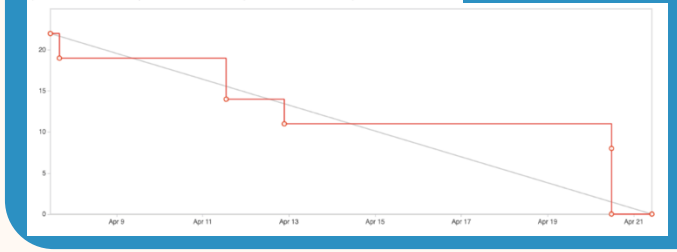
- `bridge/graph_builder.py`
- `bridge/test_graph_builder.py`

ONENOTE

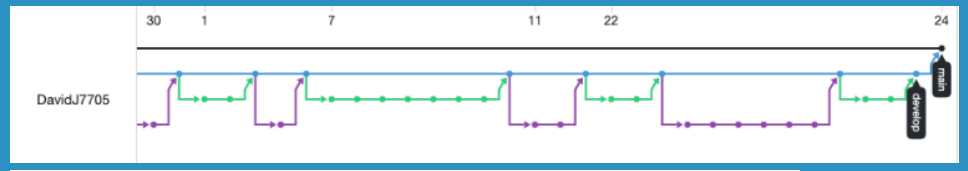


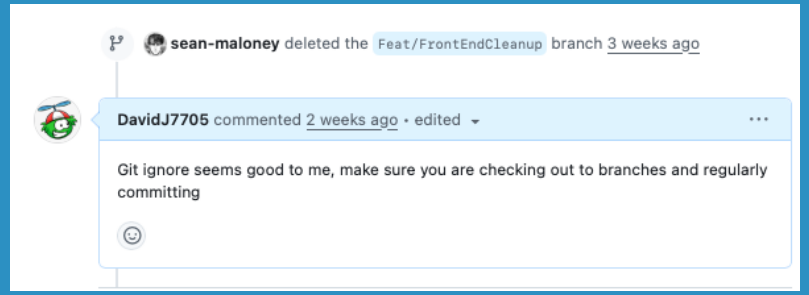
RETRO & PERFORMANCE



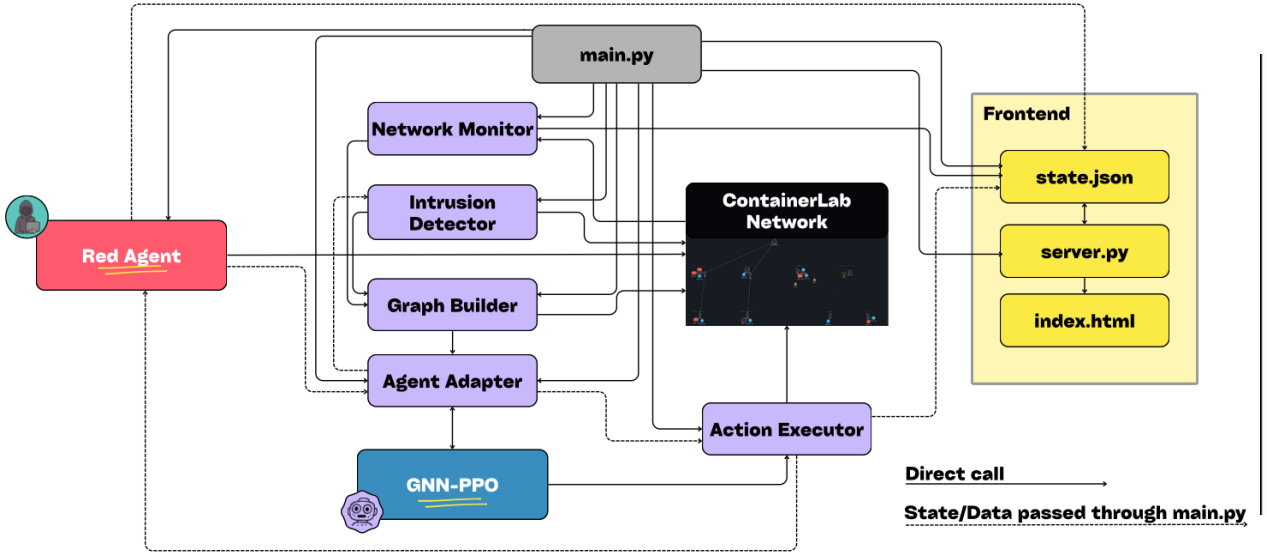


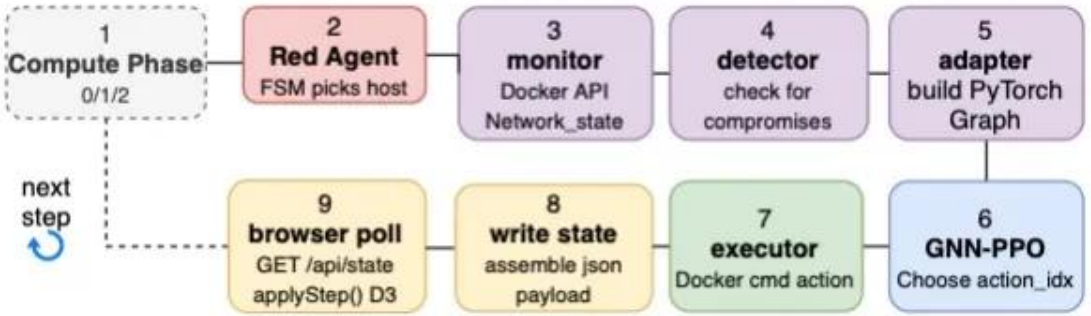
GITHUB





ARCHITECTURE





Implementation & Testing

Merge pull request #58 from DavidJ7705/develop #22

```
============= 20 passed, 1 skipped, 1 warning in 194.59s (0:03:14) =============
Name                                                                                                  Stmts   Miss  Cover
-----------------------------------------------------------------------------------------------------------------------
/home/runner/work/Network_Defender_FYP/Network_Defender_FYP/trained-agent/models/cage4.py               247     81    67%
/home/runner/work/Network_Defender_FYP/Network_Defender_FYP/trained-agent/models/memory_buffer.py        52     31    40%
/home/runner/work/Network_Defender_FYP/Network_Defender_FYP/trained-agent/models/utils.py                33     30     9%
action_executor.py                                                                                      177     86    51%
agent_adapter.py                                                                                         42      6    86%
graph_builder.py                                                                                        106     14    87%
intrusion_detector.py                                                                                    39      6    85%
network_monitor.py                                                                                       79     13    84%
red_agent.py                                                                                            105     22    79%
tests/conftest.py                                                                                         4      0   100%
tests/test_action_executor.py                                                                            45      8    82%
tests/test_agent_adapter.py                                                                              23      4    83%
tests/test_graph_builder.py                                                                              40      1    98%
tests/test_intrusion_detector.py                                                                         49      0   100%
tests/test_monitor.py                                                                                    23      0   100%
tests/test_pipeline.py                                                                                   22      1    95%
tests/test_red_agent.py                                                                                  31      0   100%
-----------------------------------------------------------------------------------------------------------------------
TOTAL                                                                                                  1117    303    73%
```

docker-compose.test.yml

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```yaml
networks:
  clab-test:
    name: clab-test

services:
  internet-router:
    container_name: clab-cage4-defense-network-internet-router
    image: alpine:latest
    command: sleep infinity
    networks: [clab-test]

  restricted-zone-a-router:
    container_name: clab-cage4-defense-network-restricted-zone-a-router
    image: alpine:latest
    command: sleep infinity
    networks: [clab-test]

  restricted-zone-a-server-0:
    container_name: clab-cage4-defense-network-restricted-zone-a-server-0
    image: alpine:latest
    command: sleep infinity
    networks: [clab-test]
```

Challenges

Sim-to-Real Gap

- 20% Actions Wasted
- Sample()
- Research Papers – Huang & Ontañón 2022 and Wilson et al. 2024.

Root Compromises

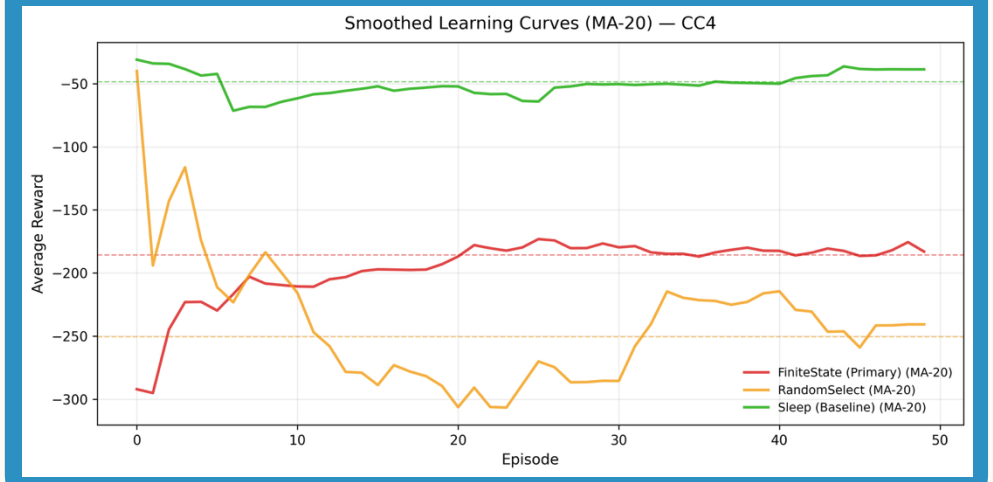
- User-level only
- Restore same host
- Match CybORG Re-imaging

Train from Scratch

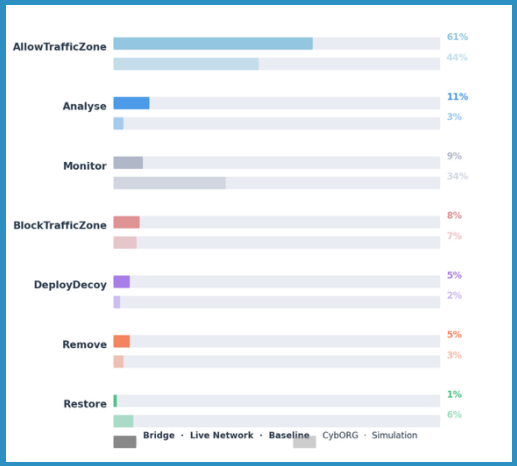
- Attempted to train PPO agent
- Ran 88 minutes with -3.5k reward
- Inference pivot due to limitations

Results

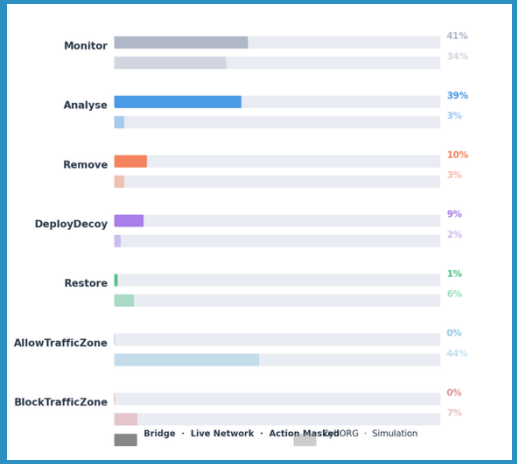
CybORG – Sim Results



BRIDGE v Simulation Action Distribution

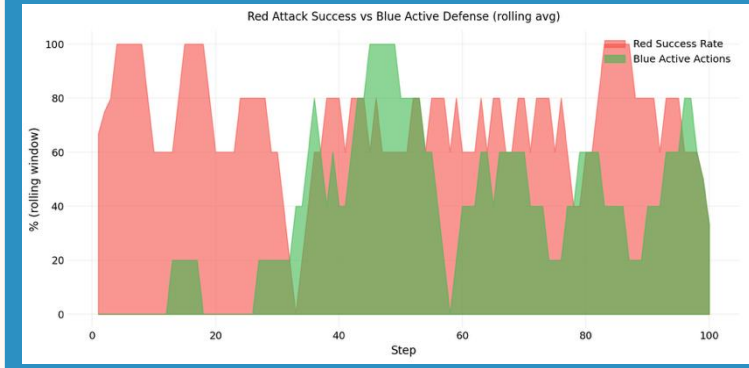


UNMASKED



MASKED

RED v BLUE Success Rate



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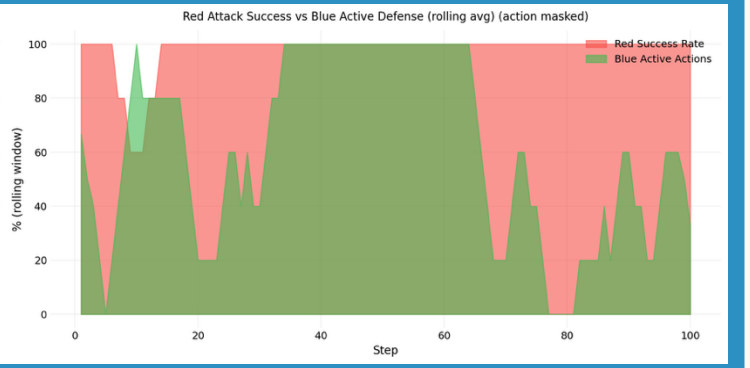

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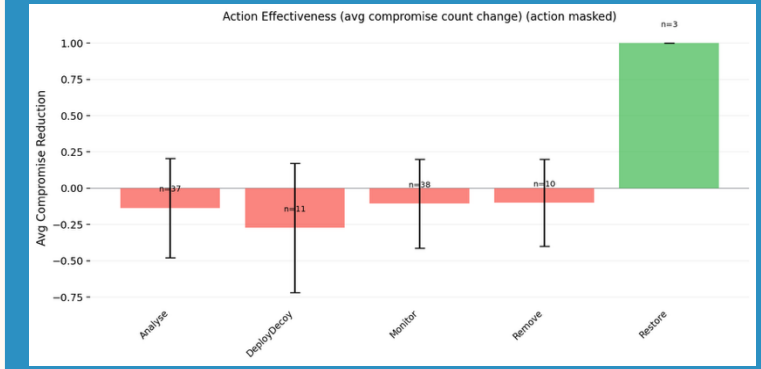
Action Effectiveness



MASKED

Conclusion

Policy Transferred

- Action distribution mirrors simulation closely

Mask Unlocked Defense

- Edge ops absorbed 69% of actions

Sim-to-Real Deployment Achieved

- Project Completed & exceeded initial plans!!